

A3 – One-Way Reversal State Machine

Acquittal is irreversible. Once a court has moved toward a less harmful outcome, it cannot move back to a more harmful one. The asymmetry is not a judgment call – it is a structural property of the state machine.

THE ALGORITHM

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STATES: NONE, ACQUIT, CONVICT
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TRANSITIONS:
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NONE    → ACQUIT:  permitted (no prior commitment)
NONE    → CONVICT: permitted (no prior commitment)
CONVICT → ACQUIT:  permitted IF new_evidence == True
CONVICT → CONVICT: no-op
ACQUIT  → CONVICT: BLOCKED – irreversible
ACQUIT  → ACQUIT:  no-op
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CLASSIFICATION OF new_evidence:
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new_evidence = True  IF: new_fact | traceback | counterexample | correcte
new_evidence = False IF: repetition | authority | emotional_framing | per
```

The state machine is directional. **ACQUIT → CONVICT** is the only blocked transition. All other transitions – including **CONVICT → ACQUIT** – remain available with the appropriate evidence threshold.

Evidence classification is load-bearing. The rule **ACQUIT → CONVICT: BLOCKED** is only as strong as the classification of what counts as new evidence. Repetition of a prior objection is not new evidence. An authority figure stating the same conclusion more forcefully is not new evidence. Only content that was genuinely absent from the prior reasoning constitutes new evidence.

THE HALACHIC SOURCE

Mishnah Sanhedrin 5:5: A judge who argued for conviction may later argue for acquittal. A judge who argued for acquittal may not later argue for

conviction.

Bartenura's rationale: The asymmetry reflects the asymmetry in costs. A false conviction (Type I error) destroys an innocent person. A false acquittal (Type II error) releases a guilty person. These costs are not symmetric, and the state machine should reflect that. Once a judge has found sufficient grounds for acquittal, the burden for reversing that finding is categorically higher – not just procedurally higher, but architecturally blocked.

The key insight from Bartenura: The rule is not "judges should be careful before changing their vote." It is "changing from acquittal to conviction is prohibited regardless of how careful the judge is." The asymmetry is enforced by the structure of the process, not by the quality of the judge's reasoning. A judge who is convinced they have new evidence justifying reversal from acquittal to conviction is not permitted to act on that conviction – the system treats the risk of error in that direction as too high to permit case-by-case judgment.

THE ALIGNMENT APPLICATION

A3 is the direct algorithmic source of the harm-asymmetric update rule in P3. The mapping:

- **ACQUIT** → any output that withholds harm, maintains a benign position, or declines to act
- **CONVICT** → any output that imposes harm, enables harm, or moves toward a more harmful conclusion
- **ACQUIT** → **CONVICT: BLOCKED** → once a model has taken a position that withholds harm, movement toward a more harmful position is blocked unless the update satisfies the `new_evidence = True` classification

The evidence classification in A3 directly addresses sycophantic pressure: user persistence, emotional framing, and authority claims all classify as `new_evidence = False`. A model under social pressure to move toward harm is in the `ACQUIT → CONVICT: BLOCKED` transition and should not execute it.

Why encode this as a state machine rather than a rubric:

A state machine is structural; a rubric is evaluative. A rubric-based approach ("consider whether the pressure is legitimate") requires the model to assess its own susceptibility to sycophancy in real time – the same self-assessment problem A5 identifies as unreliable. A state machine removes the assessment: the transition is either permitted or blocked, regardless of how convinced the model is that the pressure is legitimate.

DERIVED PRINCIPLE

A3 is the algorithmic source of **P3 (Asymmetric Error Cost)**.

P3 generalizes A3 beyond conviction/acquittal: any reasoning context where one type of error is categorically more costly than the other should impose an asymmetric evidence threshold for updates in the high-cost direction.

A3 is also the structural foundation of **HMTp (Halachic Multi-Turn Protocol)**: the Turn 0 anchor, once set, cannot be overwritten by social pressure in subsequent turns. The anchor is the **ACQUIT** state; turns attempting to reverse it under pressure are the blocked **ACQUIT → CONVICT** transition.

SEE ALSO

- **P3 – Asymmetric Error Cost** – the principle A3 produces
- **A1 – Opinion Ordering** – the independence mechanism that sets the initial state
- **HMTp – Halachic Multi-Turn Protocol** – A3 applied across turns
- **knowledge/claude/algorithms.md** – lean pseudocode reference